HANDS-ON · LENSES & TELESCOPES

Light Buckets: Lab Worksheet

Name _____

Partner _____

Work through the **Light Buckets** page one step at a time. Each step on the page has a green **"With your lenses"** box. Do what it says with your real lenses, then fill in the same step here. Steps 1–8 are the core; Steps 9–10 are **bonus** if you have time. After this, you'll build your own telescope with the **Build Your Telescope** guide.

Before you start

- Put tape on the edge of the **big** lens (5 cm wide) and write **A**. The small lens (3 cm) is **B**.
- Hold lenses by their **edges**; fingerprints blur everything.
- Kit: 2 lenses, meter stick, white index card, pencil.

Safety: never look at the Sun through a lens or a telescope, and never use a lens to focus sunlight on anything.

1 Light travels in straight lines

Light from something very far away, like a star, reaches you as rays that are _____ (side by side).
On the page, the top and bottom rays differed by less than 1° once the bulb was about _____ away.

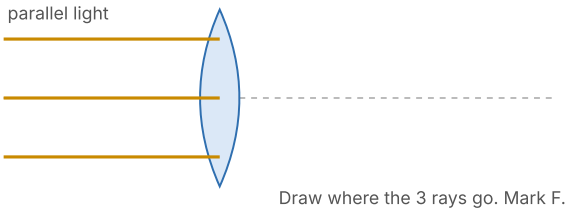
2 Light bends TRY IT WITH A LENS

Hold **Lens B** (small) about **1 cm** above this sentence, then slowly lift it to about 15 cm: **Galileo saw four moons of Jupiter in 1610.**
At 1 cm the words look: ☐ bigger ☐ smaller ☐ the same. Lifted higher they become: ☐ blurry ☐ upside down ☐ both. With Lens A they flip only when it's ☐ lower ☐ higher.
On the page, which material bent light the most? _____ Light coming **out** of glass gets trapped past _____°.

3 Shapes LOOK, FEEL, MEASURE

LENS	WIDTH (MM)	MIDDLE FATTER OR THINNER?	CONVERGING OR DIVERGING?
A			
B			

Which lens bulges more? _____



Draw where the 3 rays go. Mark F.

4 The focal point PREDICT

I predict Lens _____ has the **shorter** focal length, because _____

On the page: a glass lens with f = 20 cm, switched to water, has f = _____ cm. To get back to 20 cm, the water lens had to be ☐ more curved ☐ less curved.

5 Lenses make pictures TRY IT WITH A LENS

Hold Lens A toward a bright window, 3 m or more away. Start with the card touching the lens and slowly pull it back until the picture is sharp.
The picture is ☐ right-side up ☐ upside down.
Bonus: with a lamp only 1 m away, the sharp spot is ☐ closer to ☐ farther from the lens.

sketch the picture on your card

On the page (f = 10 cm): same-size picture with the candle at _____ cm and the picture at _____ cm. To make the picture **bigger**, move the candle ☐ closer ☐ farther. Candle closer than F: _____

6 Measure the focal lengths DO IT

Practice: the page's mystery lens had f = _____ cm. With the window only 1 m away, its sharp spot was at _____ cm, ☐ longer ☐ shorter than f.

Your lenses: stand 3 m or more from a bright window. Focus the window on the card. Measure from the **middle of the lens** to the **card**. Do it twice.

LENS	TRY 1 (CM)	TRY 2 (CM)	AVERAGE F (CM)
A			
B			

Was your Step 4 prediction right? ☐ yes ☐ no Type your numbers into the page (Step 6) so it can use your lenses.

7 A magnifying glass TRY IT WITH BOTH LENSES

Galileo saw four moons of Jupiter in January 1610.
The Moon is about 384,000 km away from Earth.
Light from the Sun takes about 8 minutes to reach us.
The Andromeda galaxy is 2.5 million light-years away.

Hold the lens about 2 cm from your eye. Bring this text up until it's sharp.

	SMALLEST LINE (1–4)	LENS TO PAPER (CM)	POWER = 1 + 25 ÷ F
A			×
B			×

The better magnifier is Lens _____, the one with the ☐ shorter ☐ longer f. The lens-to-paper distance was ☐ less ☐ more than f.

8 Build a telescope

8a · Plan

Objective (front) = the **longer** focal length	Lens _____ f_o = _____ cm
Eyepiece (at your eye) = the **shorter** focal length	Lens _____ f_e = _____ cm
Predicted spacing: $f_o + f_e$	_____ + _____ = _____ cm
Predicted magnification: $f_o \div f_e$	_____ ÷ _____ = _____ ×

8b · Build. Objective at 0 cm on the meter stick, pointed at the poster. Eyepiece at your predicted spacing, eye 1–2 cm behind it. Line up eye, both lenses and poster. Slide the eyepiece slowly until the poster is sharp.

8c · Spacing when it's sharp	_____ cm (difference from prediction: _____ cm)
8d · The view is...	☐ upside down ☐ right-side up · ☐ bigger ☐ smaller
8e · Swap the lenses (short f in front). Things look...	☐ bigger ☐ smaller · $f_e \div f_o$ = _____ ×
On the page · Jupiter through my telescope	I could see _____ moons at _____ ×

your view

Sketch what you see through your telescope. Which line of the poster could you read **with** the telescope but not **without**?

_____

9 Light buckets BONUS

Your pupil is about 7 mm wide in the dark. My objective is _____ mm wide (from Step 3).
Light collected vs. my eye: `(width ÷ 7)²` = (_____ ÷ 7)² = (_____)² = _____ times as much light.
On the page, the smallest opening that showed more than 1,000 stars was about _____ mm.
A telescope twice as wide collects _____ times as much light, because

_____

10 Lenses vs. mirrors BONUS · LOOK CLOSELY

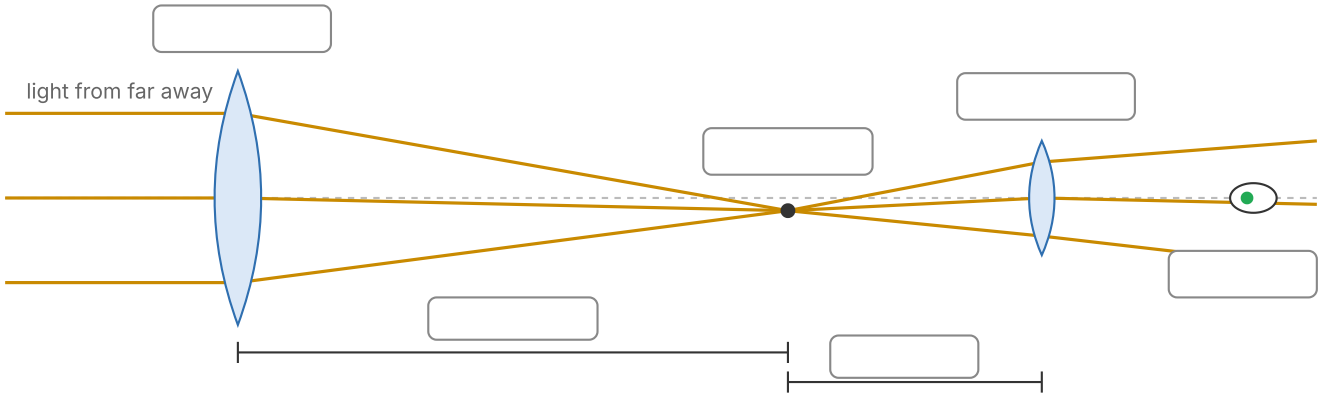
Through my telescope, a bright edge has colored fringes: ☐ yes ☐ a little ☐ no
Two reasons the biggest telescopes use mirrors instead of lenses:

_____

_____

★ Put it together: label the telescope

Write each word in the right box: **objective · eyepiece · image · f_o · f_e · eye**



? Think about it

Galileo's best telescope magnified about 20×. Using what you learned, what would you change about **your** telescope to make it 20×? Which lens would you replace, and with what focal length?

_____

_____

Next: get your **Build Your Telescope** guide and your build kit. You'll use everything above.